Overview

Team 7
Instructor Name: Dr. Sopha
Section Number: 26782
Names:

- Rithik Ganesan
- Michael Gant
- Brinton Nelson
- Humberto Noriega
- Chloe Sanborn

Emails:

- rkganesa@asu.edu
- msgant@asu.edu
- bnelso56@asu.edu
- hnorieg4@asu.edu
- csanbor2@asu.edu

Link to Youtube Presentation: https://youtu.be/_cEalPchW-g

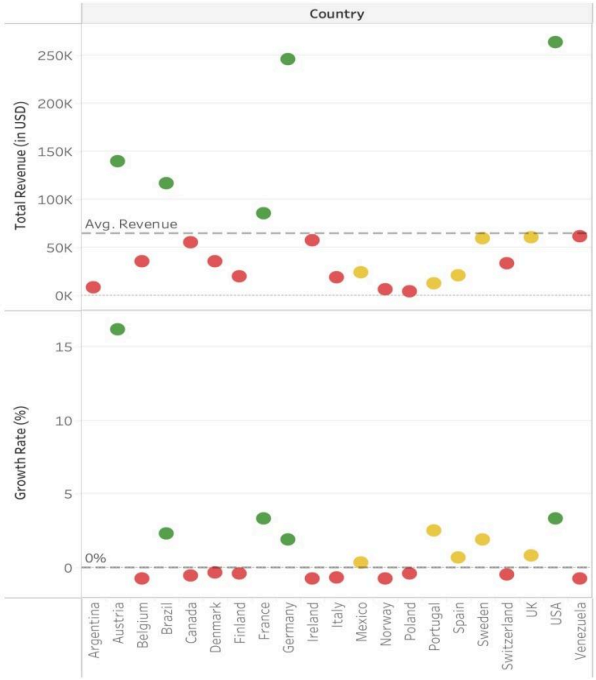
Executive Summary:

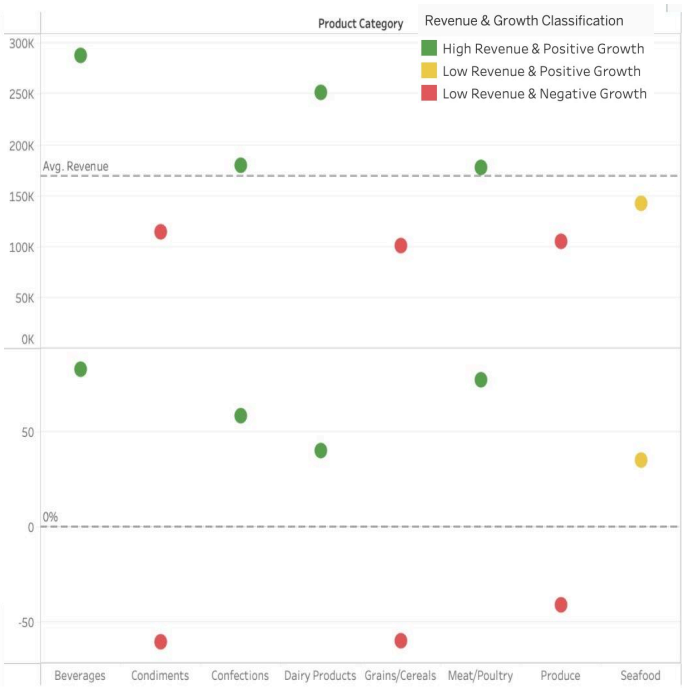
This analysis of Northwind Trader's global sales data highlights key areas where the company is excelling, along with potential growth opportunities. Our team had many key findings. Northwind's strongest revenue drivers are mostly located in Europe and the USA contributing over 60% and 20% of total revenue respectively. Latin America representation is limited along with no business generated in Asia and Africa. Beverages and Dairy are our strongest product categories contributing over 33% of revenue. The other product categories do not contribute as much to Northwind's revenue but some of these categories have seen significant growth. All in all, we identified the strongest countries/categories that we should continue to expand and develop. The weakest countries/categories that Northwind should either reconsider providing to/providing as a whole or implementing new strategies to improve efficiency. As well as the countries/categories that indicate opportunities for growth.

We selected our 5 deep dive areas to align with Northwind's mantras of "Transformation" and "Run to where the ball will be." Each area was chosen to identify the strongest points of the business so that we can transform to focusing on these points as that is where the ball and the money will be.

1. Number of Orders By Country: to identify key operational areas to reduce shipping costs and efficiency in highly concentrated countries
2. Total Revenue By Country: to identify the strongest markets and opportunities for growth
3. Top Revenue Companies: to assess our top contributors and where they're located
4. Revenue By Product: to evaluate which products are worth continuous investment
5. Revenue Trends by Category: to track seasonal trends to optimize stocking and pricing

High Level Overview:

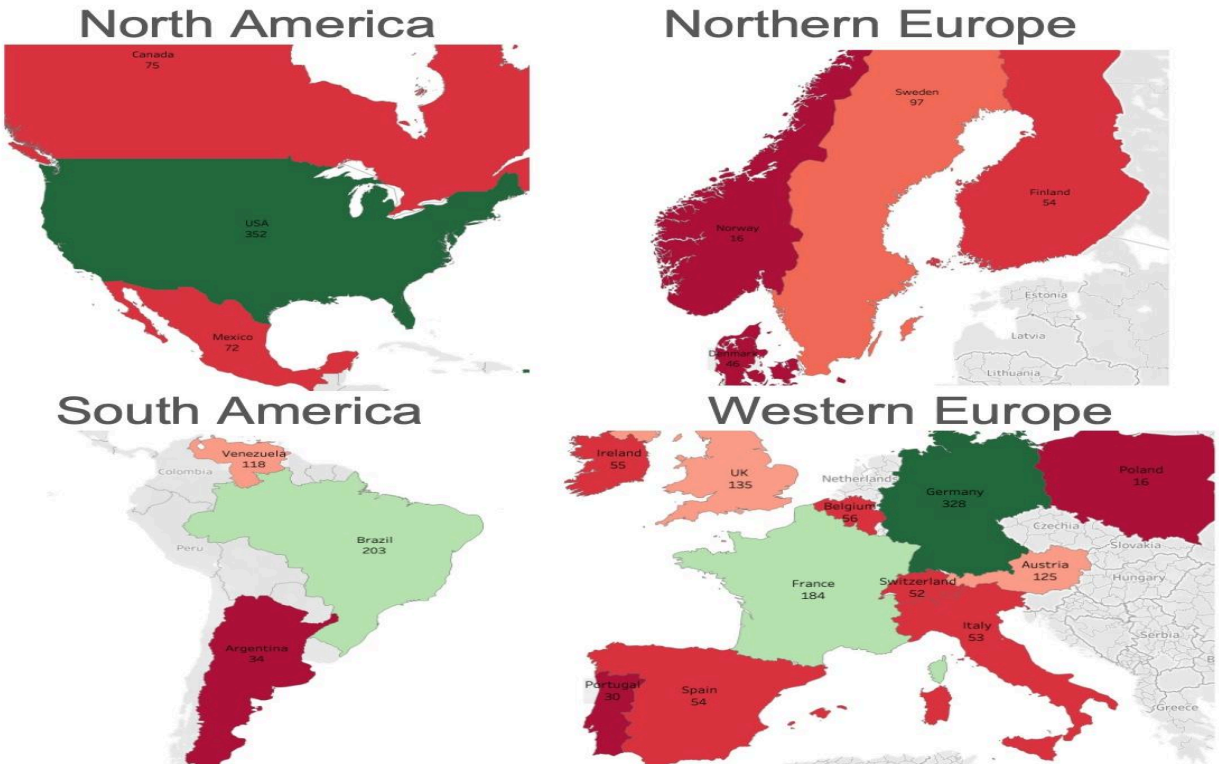




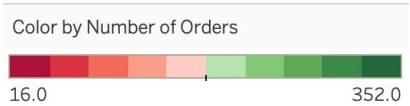
Actionable Insights:

- The countries and product categories in green have seen high revenue and growth, proving to be our strong-suits and areas we should continue to expand in.
- The yellow circles signify low revenue and positive growth. Although these areas haven't been the most profitable, they have seen growth over time and may be worth continued pursuit.
- The red circles represent low revenue and low growth areas. Northwind should either re-evaluate whether these areas are worth continual investment of resources, or investigate the causes behind the low performance in these areas.

Number of Orders by Country



Filled map to provide a geographical representation of the distribution of our orders. Clearly identifies the countries in which we have a high and low concentration of orders. This helps Northwind understand where to base and focus our operations to reduce shipping costs and increase efficiency.



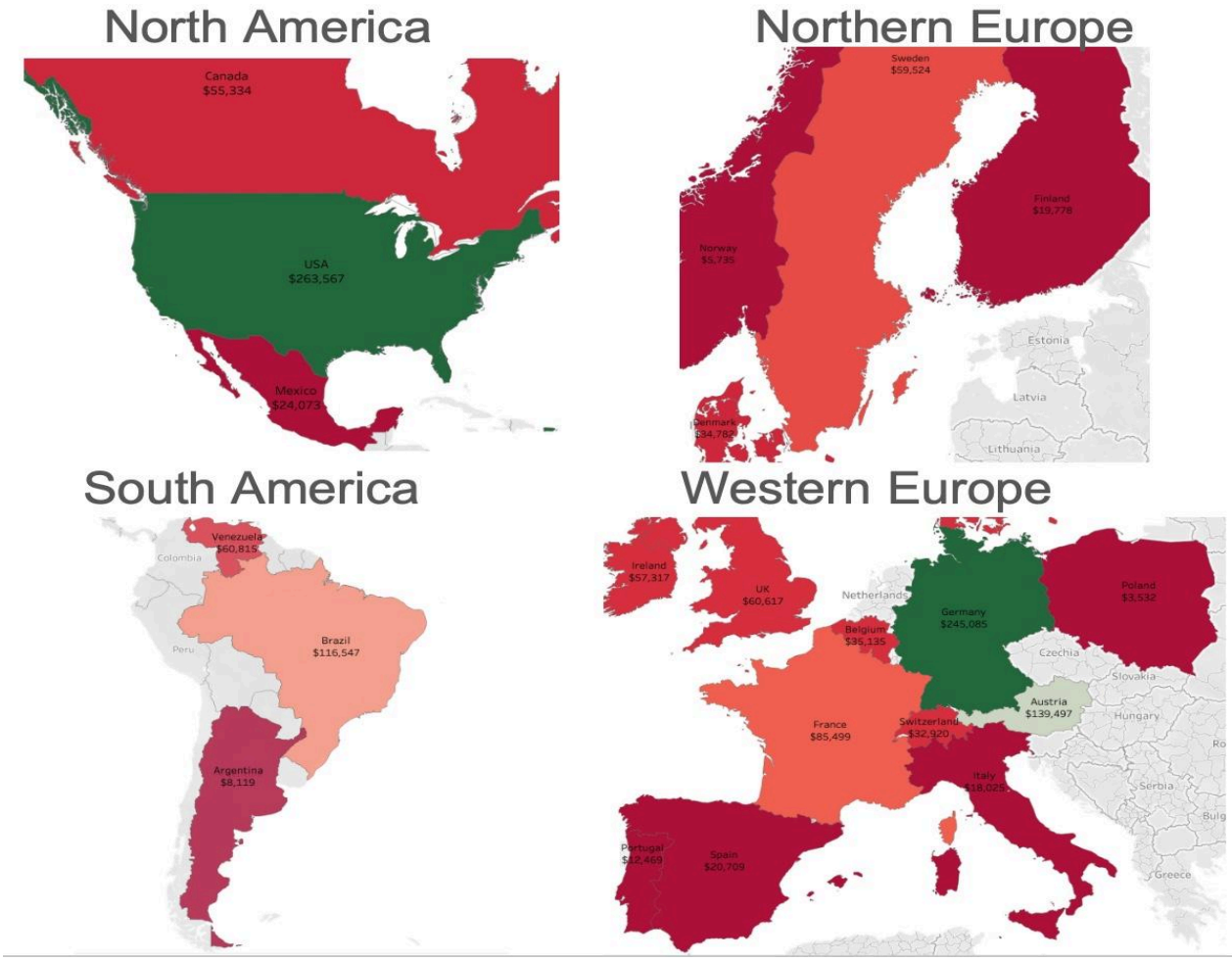
Key Insights:

- USA, Brazil, Germany, and France hold the highest amount of orders, combined, they account for just under half the total orders worldwide
- Argentina, Belgium, Norway, Poland, and Portugal have a relatively low order total

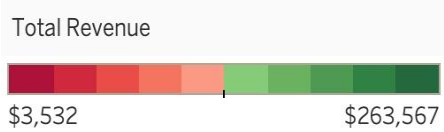
What we Don't Know:

- Marketing strategies and reasoning behind low concentration of orders in countries
- Product storage locations and the associated shipping costs for transporting products from these locations to far away countries

Total Revenue By Country (in USD)



The distribution of revenue across each country shows where the company's strongest market share is and what opportunities exist for growth and identify underperforming areas.



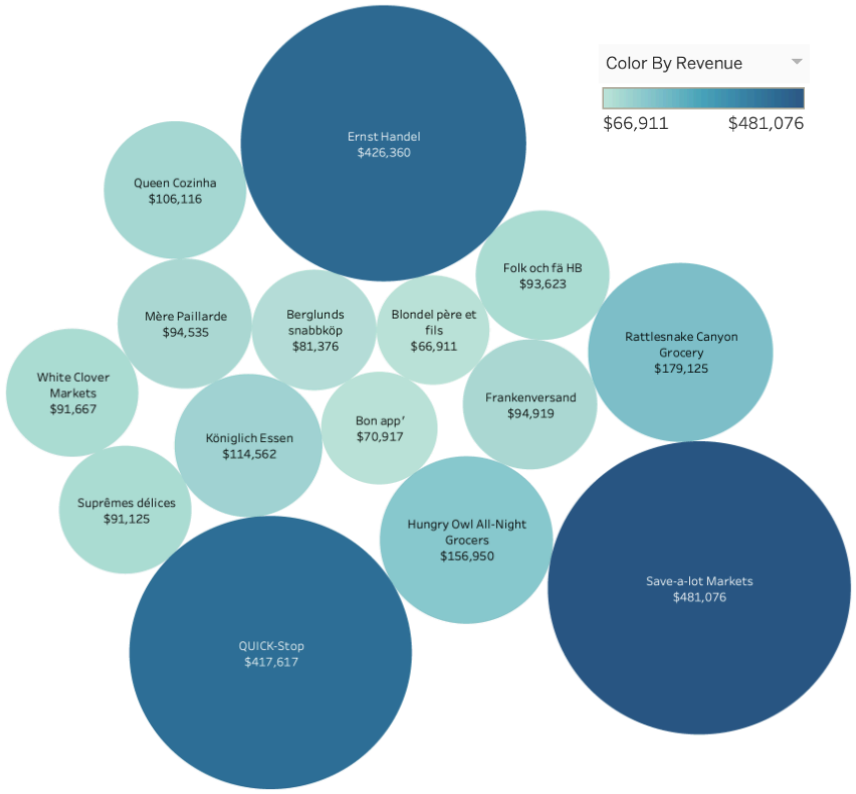
Key Insights:

- USA contributes over 20% of total revenues
- Countries in Northern Europe potentially show underperformance
- Revenue disparity between countries in Western Europe demonstrates further investigation into causality

What we Don't Know:

- What resources are being allocated to Germany that are causing such high performance when compared to other countries in the region
- Market Research that explains why similar countries of population and demographics are not having a higher market share in their respective countries

Total Revenue from Companies (Top 15 in USD)



Country	Company Count
USA	3
Germany	3
Austria	2
Sweden	2
Brazil	2
Canada	2
Ireland	1
France	2
Belgium	1
Venezuela	1
Denmark	1

The 15 companies that have given Northwind the most revenue are displayed in a bubble chart. The table is the countries that contain the top 20 companies that have given us the most revenue.

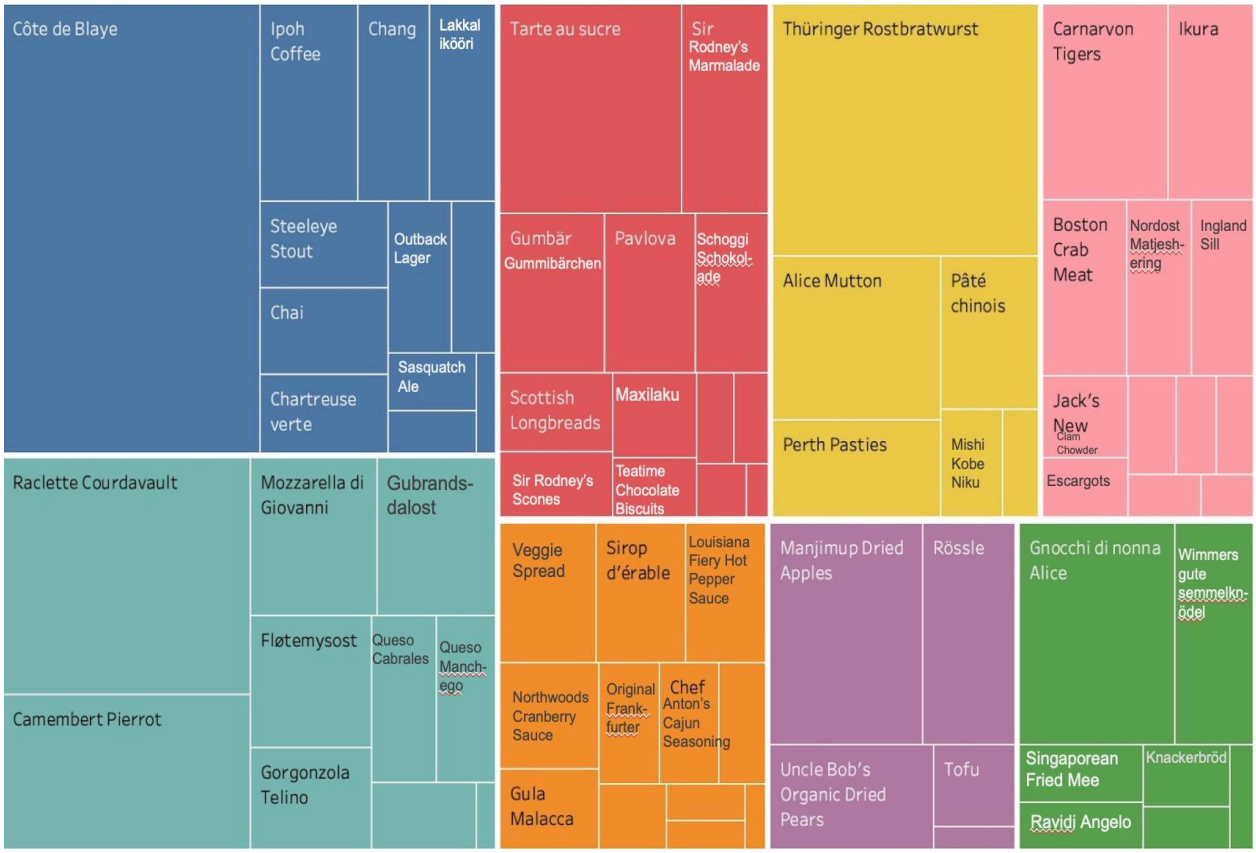
Key Insights:

- Europe dominates revenue generation, contributing over 60% of the total.
- North America is the second-largest contributor, with U.S. and Canadian companies generating substantial revenue.
- Latin American representation is limited, with three companies from Venezuela and Brazil accounting for a smaller share.
- No representation from Asia, Africa, or Oceania, suggesting untapped markets.

Strategic Applications:

- Analyze what drives high-revenue companies' success.
- Replicate effective marketing and product strategies to maximize revenue.
- Allocate resources (inventory, customer service) to strengthen relationships with top-earning companies.
- Assess whether to invest in lower-revenue companies or shift focus elsewhere.
- Use regional performance trends to guide expansion and explore partnerships with companies sharing characteristics of top earners.

Treemap of Total Revenue By Product



This treemap visualizes exactly where our revenue is coming from. It clearly shows that Beverages, Dairy Products, Confections, and Meat/Poultry are our best performing product categories.

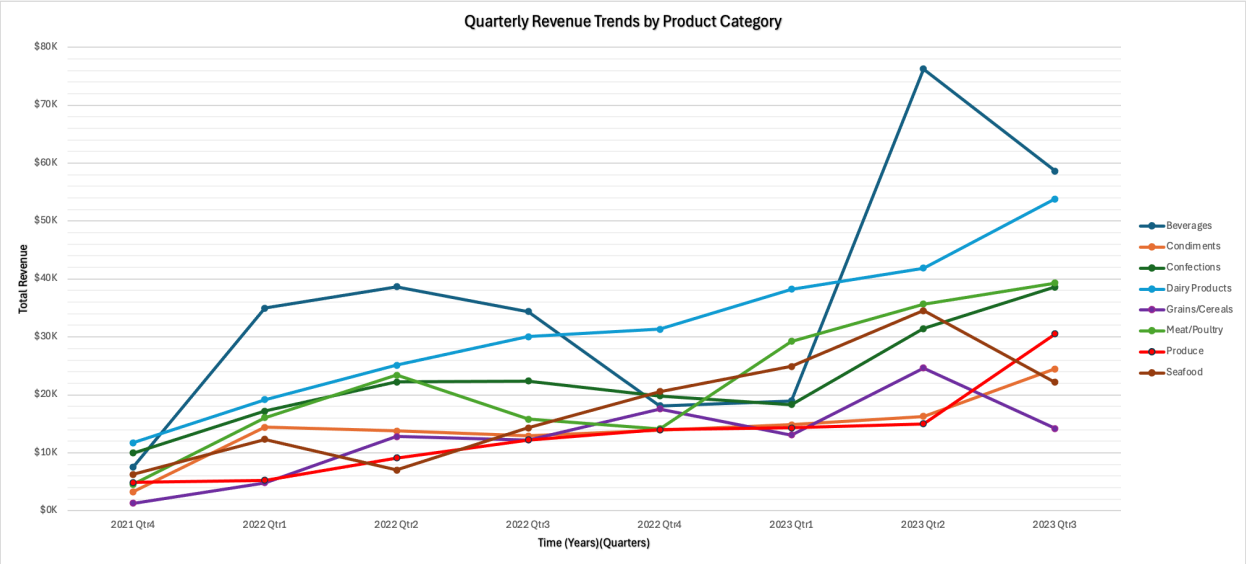
Key Insights:

- Beverages and Dairy Products are responsible for over 33% of Northwind Trader's revenue
- Confections and Meat/Poultry products also make up a notable chunk of total revenue
- Produce and Grain/Cereal products make up a relatively small portion of total revenue
- Côte De Blaye, Tarte au sucre, Thüringer Rostbratwurst, Alice Mutton, Raclette Courdavault, Camembert Pierrot, Carnarvon Tigers, Manjimup Dried Apples, and Gnocchi di nonna Alice are the products that bring the most revenue
- All product names not listed contribute a relatively low amount of revenue

What we Don't Know:

- Growth/decline of products over time
- Stock Levels, Pricing, and Discounts: these are key factors that influence sales and revenue

Quarterly Revenue Trends by Product Category (in USD)



This data highlights which product categories generate the most and least revenue by quarter, allowing for better stocking and pricing decisions to optimize costs, sales, and revenue.

Key Insights:

- Beverages experience strong spikes in Q2 and Q3 but decline in Q1 and Q4, indicating seasonal volatility
- Dairy Products show steady and reliable growth, making them a strong investment category
- Seafood revenue had been consistently increasing, yet there is a steep dropoff in the last quarter warranting further investigation
- Produce revenue has surged in the most recent quarter, suggesting increasing demand

This visualization provides valuable insights, but additional information is needed to verify the causation to ensure accurate strategic decisions.

What we Don't Know:

- Stock Levels and Discounts: these are key factors that influence sales and revenue
- Geographical Breakdown: seasonality impacts markets differently across different countries

Conclusion: The data reveals that Western Europe and the USA are crucial markets, contributing significantly to revenue, while underperformance in certain regions presents opportunities for expansion. Beverages and Dairy Products emerge as dominant revenue drivers, with seasonal fluctuations indicating the need for optimized inventory management. However, gaps remain in understanding the factors influencing regional performance, product demand fluctuations, and the role of pricing and stock levels. Further investigation into these areas will enable more precise strategic actions. To support the SLT in making informed decisions, we recommend conducting deeper market research, assessing resource allocation across regions, and analyzing historical sales trends. By leveraging these insights, Northwind Traders can enhance efficiency, strengthen key market positions, and capitalize on emerging opportunities to drive sustainable growth.

Request For Proposal

Project Overview

Northwind Traders is a leading international reseller and distributor of high-end foods and select beverages, serving grocery stores across the Americas and Europe. As part of our ongoing transformation initiative, we seek external expertise to analyze our sales data, identify actionable insights, and recommend strategies to optimize our operations and customer engagement. The company aims to increase sales, reduce costs, and have the capability to outperform competitors in a dynamic market environment.

This Request for Proposals (RFP) seeks submissions from qualified vendors who can help us utilize advanced data analytics, optimize our processes, and engage in strategic planning. The goal is to achieve significant enhancements in our profitability and operational efficiency through these initiatives.

Project Goals

The primary objectives of this project are:

1. **Sales Optimization**: Have the ability to identify high-performing and underperforming markets, products, and customer segments to reallocate resources effectively.
2. **Cost Reduction**: Analyze Activity-Based Costing (ABC) data to pinpoint inefficiencies and recommend cost-saving measures.
3. **Customer Insights**: Understand customers preferences and buying patterns to enhance targeting and retention strategies.
4. **Operational Efficiency**: Streamline supply chain and distribution processes to reduce lead times and improve service levels.
5. **Strategic Growth**: Provide actionable recommendations to expand into new markets and product categories.
6. **Unlocking Sales Growth and Efficiencies**: Identify the main factors contributing to sales growth and the courses of cost inefficiencies.
7. **Real Time Sales Performance Visualization**: Adopt advanced visualization tools to proactively monitor and enhance sales performance in real time.

Scope of Work

1. **Data Analysis**: Perform an in-depth analysis of Northwind Traders' sales, costs, and customer data utilizing advanced analytics tools to uncover actionable insights and drive strategic decision-making.
2. **Visualization and Reporting**: Develop clear, impactful visualizations (e.g., maps, trendlines, bubbles, bar charts, etc.) to communicate insights to the Senior Leadership Team (SLT).
3. **Benchmarking**: Compare Northwind Traders' performance against industry benchmarks and competitors.
4. **Strategy Development**: Propose actionable strategies to achieve the project goals, including short-term and long-term recommendations.
5. **Implementation Support**: Provide guidance on implementing the recommended strategies, including change management and training.
6. **AI-Powered Sales Trend Prediction**: Leverage advanced AI-driven forecasting models to accurately predict future sales trends and optimize decision-making for sustained growth.

7. **Sales Data Cleanup and Validation**: Evaluate and refine historical sales data to ensure its accuracy and comprehensiveness.

Current Roadblocks and Barriers to Success

1. **Data Fragmentation**: Sales and operational data are distributed across multiple regions and systems, making consolidation and analysis challenging.
2. **Limited Insights**: Current report lacks depth and actionable insights, hindering strategic decision-making.
3. **Resource Constraints**: Internal teams are stretched thin, limiting their ability to conduct in-depth analyses.
4. **Market Volatility**: Fluctuating customer demand and supply chain disruptions pose risks to achieving consistent growth.

Evaluation Metrics and Criteria

1. **Relevance of Experience**: Demonstrated expertise in data analytics, sales optimization, and cost reduction for similar industries.
2. **Methodology**: Clear and innovative approach to data analysis, visualization, and strategy development.
3. **Deliverables**: Quality and practicality of proposed deliverables, including reports, visualizations, and recommendations.
4. **Cost-Effectiveness**: Competitive pricing and alignment with Northwind Traders' budget.
5. **Timeline**: Ability to deliver results within the agreed timeframe.

Submission Requirements

1. **Company Profile**: Overview of the vendor's experience, capabilities, and relevant case studies.
2. **Proposed Approach**: Detailed Methodology and timeline for completing the project.
3. **Team Composition**: Profiles of key team members, including their qualifications and roles.
4. **Cost Proposal**: Breakdown of costs, without forgetting fees, expenses, and payment terms.
5. **References**: Contact information for at least three clients for whom similar services were provided.

Potential Recipients

1. **Tableau Professional Services**: Accomplished professionals in data visualizations and analytics, renowned for their ability to convert raw data into impactful, actionable insights that drive decision-making and strategy.
2. **Deloitte Consulting**: Known for its data-driven approach to solving complex business challenges.
3. **Google Cloud AI & Analytics**: Delivers powerful cloud-based analytics and innovative AI-driven solutions designed to maximize sales performance and drive optimal results.
4. **Oracle Analytics**: Renowned for its powerful business intelligence solutions and cutting-edge machine learning capabilities.

RFP Conclusion:
At Northwind Traders, we recognize that to thrive in today's competitive landscape, we must continually innovate and enhance our sales data capabilities. That's why we are eager to collaborate with a qualified vendor who shares our vision for growth and excellence. By partnering with us, you will not only play a pivotal role in driving informed decision-making but also help us maintain and strengthen our competitive edge in the market. Together, we can unlock the full potential of our sales data, transforming insights into actionable strategies and paving the way for mutual success. We look forward to collaborating with a vendor to being one of the most outstanding companies in the market!